



How to Create Fun Games

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1. What is Fun?
2. Rational Design - Universal Game Mechanics
3. Understanding what game you are making?



What is fun?





The fun is in trying to **win the game...**



Its like in movies, if we can anticipate what's going to happen, it's *boring*.

If you always win, with the same technique, it's boring.

We want to be challenged by the opposition.

It's not fun to constantly beat your little brother 16-0 at NHL

Unknown Outcomes

The key is keep the player in the realm of unknown outcomes.

Once a pattern is bested, Players will always desire more complex/sophisticated game system to master.

Too complex = Noise = Losing interest



That's a reason old genre are dead or dying:
You already beat them, you know how it ends before the end.

2D beat them up -> 3D open worlds = More gameplay Possibilities

You can make old gameplay interesting by making it *harder*:

2D Shoot them up became Bullet Hell

2D platformer became the <indie platformer>

Adventure games embraced Roguelike (random elements) and Permadeath (hard)

The Deck of cards is the greatest game platform of all time

Simple concept: 4 types, 13 cards per type (1-10 plus King, Queen, Jacks) plus 2 Jokers.

The secret is that you will never have two identical card deck that are identical - ever
On this premise, you can then build simple but **never ending fun games**:

Solitaire

Blackjack

Poker

Etc...

All this is possible because of the uncertain outcome of the card sequence each time we shuffle it, yet the deck is really easy to understand - Easy to learn - hard to master

The Multiplayer effect

Other players are the **best Enemy AI ever**

You ***never know*** who you are going to face, and what tactic they are going to use.

Very simple mechanics becomes a showcase of who is the fastest, who's the more precise, etc.

Giving everybody a chance to win is why there's skill based matchmaking.



Unknown Outcome is the key of Fun games



What is the Game Designers Job?

Find the Fun





The Universal Game Mechanics

The Ingredients of Challenge



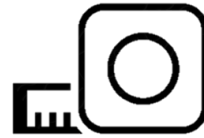
The 8 Universal Game Mechanics



Timing



Reflexes



Measurement



Strategy



Precision



Management



Tactics



Cleverness

Rational Game Design

1. Define the experience you want
2. Choose the Appropriate game mechanics
3. Generate Variety

How to identify the right mechanics

What is your game About?

Adventure?

Romance?

Action?

Start with the idea, it's going to help you later when it's time to make choices

Example with 2 game concepts

WORLD
WAR II





WW2 Romance game

Great Dialog System with branching choices,

Choice in actions. All Choices counts

Great writing with characters to fall in love with.



WW2 Action game

Great shooting mechanics

Memorable Action Set-Pieces

Vehicles (Tanks, Planes, etc)



Both games may share certain features, but if the **Romance game's** budget and time is mainly spent on developing **great shooting mechanics**, it is *no longer a romance game*.



Your lighthouse when it's time to make choices

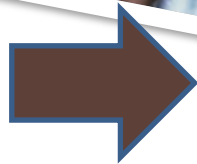
Mandate



Create a sequel to
World War II Shooter Game



World War II shooter game



World war II-2
new awesome feature:

AI Partner!

What is the game?

3rd Person Shooter

Objective: Kill the enemy

Challenge on

- Precision
- Reflexes
- Tactics

AI Companion - What does it do in this game?

+++

- New *tactics* in combat (flank, aggro)

+++

- Player can switch between the 2 characters (with different play style)

+++

- AI can drive a car while I shoot

+

- Tell jokes during the game

+

- Solve location-based puzzle (standing on 2 plates)

- ... Use this method to prioritize your features and budget

Finally, go *deep* with your
core gameplay systems

Test Case: Gun Gameplay



Gun Gameplay RGD:

- Objective: Get Target HP to [0]
- Precision Gameplay: Align Pixel on Pixel
- Identify weak zone (ex: Head Shot)
- Enemy moves a lot or little

Example: Weapon Basic design



- [X] **Damage** per bullet (in HP)
- [X] Damage **Range Drop** (multiple zones)
- [X] bullets per seconds: **Rate of Fire**
- [X] bullets per mag: **Magazine Size**
- [X] Sec: **Reload Time**
- **Bullet Spread** multiplier: Radius of [X] within a bullet will randomly land. This Radius grows per bullet when firing more then [X] bullets in a row
- **Max Bullet Spread** (max radius)
- **Fire rate** (Single, burst, Full Auto)

[illegible]

- Grip = Spread modifier (x0.5)
- Custom Ammo = Damage multiplier (x1.2)
- Scope = Camera Zoom when scoped (x2)
- Ext Mag = More bullets in Mag
- Etc...

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Variety through Game Design

Your Variables enables a multitude of weapons with difference Strength and Weaknesses,

Affects the core gameplay (Precision) and create secondary gameplay (ex: Movement)

- Big Damage, but lots of recoil (favors short burst)
- Small Damage, but can have a Silencer
- Small Damage, no recoil, but High Rate of Fire and short range (ex: Uzi)

Variety through Level Design



Then, Level Design can use the to create interesting gameplay scenarios

- Ex: in close combat, [X] gun is better, while long range it encourage to switch to [Y].
- To stay Stealth, you chose a certain attachment (Silenced weapon), but it reduce the damage.
- Change what weapons the AI is using

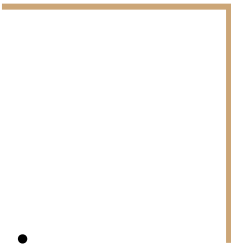
No 2 levels or encounters should be using the same pattern in your game

Add Multiplayer to the mix...

- The Unknown are Players
- Streamline the possibilities by attaching gameplay variables to 'Heroes'.
- New Heroes create new Meta



Conclusion

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1. Fun comes from Unknown Outcomes, design your game around this
 2. Use RGD to breakdown the gameplay mechanics you want use
 3. Decide what game you want to make